NutriVision: AI-Powered Food Recognition and Nutrient Estimation

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***Abstract*—This paper presents an AI-powered food intake and diet monitoring system designed to provide accurate, automated, and personalized dietary assessment by integrating multi-modal input processing with modern artificial intelligence techniques. The proposed system enables users to log their meals either by uploading food images or by providing voice-based meal descriptions through an intuitive web-based interface, reducing the dependency on manual food entry. Deep learning models are utilized for food image classification, speech-to-text conversion, and natural language processing to identify food items, portion-related information, and contextual details from user inputs. Computer vision techniques analyze food images to recognize single and mixed dishes, while speech recognition and NLP modules extract relevant meal information from spoken descriptions. To ensure reliable nutritional assessment, the identified food items are mapped to standardized entries in the USDA FoodData Central database, from which nutritional values such as calories, carbohydrates, proteins, and fats are retrieved. By combining image-based recognition, voice-based interaction, and textual analysis, the system supports flexible meal logging and improves accuracy in dietary tracking. The proposed architecture is scalable, allowing seamless integration of advanced models and large nutritional datasets, and is suitable for real-time health monitoring applications. This system can assist individuals in managing daily dietary intake, support fitness and wellness goals, and provide valuable insights for long-term health management. Overall, the proposed solution demonstrates the potential of integrating computer vision, speech recognition, and natural language processing to build an intelligent and user-centric diet monitoring platform.**

***Index Terms*—Food recognition, Nutrient estimation, Artificial intelligence, Computer vision, Natural language processing**

I. Introduction

In recent years, rapid advancements in digital technology and artificial intelligence have significantly transformed the way individuals manage their daily activities, including health and wellness. Maintaining a healthy lifestyle has become both a priority and a challenge due to busy schedules, increased consumption of processed foods, and limited awareness of nutritional values. Proper diet management plays a crucial role in ensuring physical well-being, preventing lifestyle-related diseases such as obesity and diabetes, and achieving long-term fitness goals.

Despite the availability of various diet-tracking and calorie-counting applications, many users struggle to consistently and accurately record their food intake. Traditional dietary tracking methods rely heavily on manual data entry and self-reporting, which can be time-consuming, inconvenient, and prone to inaccuracies. Users often find it difficult to estimate portion sizes or identify exact nutritional content, leading to unreliable dietary assessments and reduced motivation to continue tracking.

To address these limitations, **NutriVision** proposes an intelligent and automated dietary tracking system that leverages Artificial Intelligence (AI) techniques from computer vision, speech recognition, and natural language processing (NLP). The system allows users to upload food images or provide voice-based descriptions of their meals. These inputs are processed using deep learning models to identify food items and estimate portion sizes with minimal user intervention.

NutriVision retrieves accurate nutritional information such as calories, carbohydrates, proteins, fats, and other essential macronutrients from the *USDA FoodData Central* database. By combining multiple input modalities, the proposed system improves usability while maintaining accuracy. Additionally, NutriVision analyzes dietary patterns over time to provide meaningful insights such as calorie trends, nutrient imbalances, and unhealthy eating habits.

Furthermore, the platform emphasizes personalization and user engagement by offering features such as dietary goal tracking, progress visualization, and meal recommendations. These features encourage long-term adherence to healthy eating habits. Thus, NutriVision acts not only as a food tracking application but also as a comprehensive AI-driven dietary companion that supports users in achieving and maintaining a balanced and healthy lifestyle.

II. Objectives of the Project

The primary objective of the *NutriVision* project is to develop an intelligent and automated dietary monitoring system that significantly reduces dependency on manual food logging and minimizes inaccuracies associated with conventional diet-tracking methods. By leveraging advanced Artificial Intelligence techniques, the system aims to simplify food intake recording while enhancing accuracy, efficiency, and overall user convenience in daily dietary monitoring.

A key objective of the project is to enable seamless multi-modal user interaction by supporting both image-based and voice-based inputs. The system employs computer vision techniques, including convolutional neural networks (CNNs), to automatically recognize food items from meal images and estimate portion sizes with minimal user intervention. In parallel, speech-to-text processing and Natural Language Processing (NLP) allow users to describe meals verbally, improving accessibility for users who may face difficulties with text-based input or visual limitations.

The project further aims to apply NLP techniques to extract relevant food-related entities such as food names, quantities, and portion-related details from textual or voice-derived input. Accurate extraction of this information is essential for reliable nutritional analysis and supports the generation of personalized dietary recommendations aligned with individual dietary preferences, health objectives, and specific nutritional needs.

Another important objective of *NutriVision* is to provide precise, consistent, and standardized nutritional information through integration with the USDA FoodData Central database. The system dynamically calculates calories, macronutrients, vitamins, minerals, and other essential nutritional metrics based on identified food items and estimated portion sizes, enabling realistic dietary assessment and informed decision-making.

Additionally, the project focuses on longitudinal analysis of users' dietary patterns to generate actionable insights such as calorie intake trends, nutrient deficiencies, and identification of unhealthy eating behaviors. By presenting these insights through a responsive and user-friendly web-based interface, the system encourages informed dietary choices, sustained behavioral improvements, and long-term adherence to healthy eating practices.

Finally, the *NutriVision* project aspires to be a scalable and adaptable platform capable of integrating additional food datasets, supporting real-time health analytics, and enabling cross-platform deployment across web and mobile environments. Collectively, these objectives aim to bridge the gap between modern AI technologies and practical, everyday nutrition tracking, making personalized dietary monitoring more accurate, accessible, and engaging.

III. Methodology

The proposed *NutriVision* system follows a structured AI-driven methodology that integrates multi-modal input processing for accurate food recognition and nutritional analysis. The overall system workflow is illustrated in Fig. 1 and is designed to efficiently handle image, speech, and text-based inputs.

A. Image Processing

Users upload meal images through a web-based interface, which are preprocessed using resizing, normalization, and noise reduction. A Convolutional Neural Network (CNN), fine-tuned through transfer learning on the *Food-101* dataset, is used to classify food items and estimate portion sizes based on visual features such as color, texture, and spatial information.

B. Speech Processing

The system supports voice-based input using speech-to-text models such as OpenAI Whisper. Audio inputs are converted into textual meal descriptions, improving accessibility and allowing users to interact with the system conveniently across different languages and accents.

C. Natural Language Processing

Text obtained from user input or speech recognition is analyzed using Natural Language Processing techniques to extract food names, quantities, and portion-related details. This enables accurate interpretation of meals described in natural or informal language.

D. Nutritional Data Retrieval

Identified food items and portion sizes are mapped to the *USDA FoodData Central* database to retrieve nutritional values, including calories and macronutrients. Nutritional estimates are adjusted dynamically based on portion size to reflect actual food intake.

E. User Interface

The processed nutritional information is displayed through an interactive web-based interface, allowing users to monitor daily intake, analyze trends, and review dietary history. The interface is designed to be clear, responsive, and user-friendly to support effective dietary monitoring.

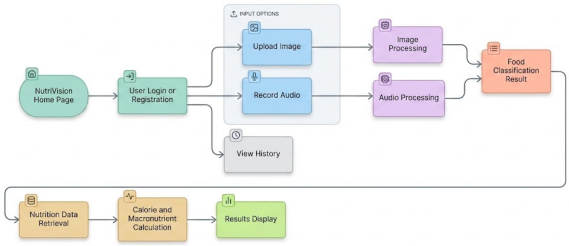


Fig. 1: Flowchart of NutriVision system workflow.

IV. Literature Survey

Ref	Authors	Year	Key Contributions	Limitations
[1]	Jenice Bhavsar; Satvik Khara; Jay Dave	2025	1. AI-based personalized diet planning 2. Food recognition systems 3. Disease prediction models 4. Improved tracking accuracy 5. Supports public health interventions	1. Poor data quality 2. Algorithmic bias 3. Ethical and privacy concerns
[2]	Gwo Giun Chris Lee; Wen-Harn Pan; Hsin-Chuan Wu; Chin-Sung Chang	2025	1. 18,737 food images dataset 2. 93 food categories 3. Focus on Asian diets 4. Supports ingredient recognition 5. Useful for health applications	1. No accuracy results 2. Single-ingredient focus 3. Limited for complex dishes
[3]	Qinpei Zhao; Weixiong Rao; Yi Wang; Xinyuan Lu	2024	1. Nutrition estimation for Chinese food 2. DChiFood dataset (220K+ images) 3. Segmentation and recognition 4. Volume estimation for nutrients	1. Low accuracy (37.5% top-1) 2. Visually similar foods
[4]	Bharanishree V; S. Mohammed Mansoor Roomi; S. Rajaram	2025	1. Review of deep learning techniques 2. CNNs, RNNs, modern models 3. Identifies key research challenges 4. Future research directions	1. Limited dataset diversity 2. Low interpretability 3. High computational cost 4. Real-time deployment challenges
[5]	Amit Kumar Nandanwar; Mohit Kushwaha; Dhirendra Pratap Singh	2024	1. Indian food nutrition system 2. CNN-based recognition (91.87%) 3. Real-time calorie tracking 4. Nutrient database mapping	1. No portion estimation 2. Manual database
[6]	Thessalonica Maria Luisse Rosaline; Calvin Anacia Suciawan	2024	1. YOLOv8 ingredient detection 2. Calorie estimation 3. Node.js + MySQL integration 4. Personalized recipe generation	1. Limited dataset 2. Image-quality dependent 3. Limited real-time performance
[7]	Wei Zuo; Weiwei Zhang; Zengchao Ren	2023	1. PReNet model using PyTorch 2. Food2K dataset 3. Progressive training 4. Self-attention mechanism	1. High computational cost 2. Dataset bias 3. Single platform images
[8]	Shih-Chuan Huang; Wei-Chun Chiang; Ya-Ting Yang	2023	1. Raspberry Pi based system 2. Depth camera usage 3. Portion-size estimation 4. Nutrition analysis in trays	1. Hardware dependent 2. Limited general use 3. Not smartphone-friendly
[9]	Amay Jaiswal; Abhay Solanki; Anshu Tomar; Shreya Bhaduria	2023	1. Diet Recall mobile app 2. KNN-based recommendation 3. Content-based filtering 4. Fitness tracking integration	1. Manual logging 2. Basic KNN model 3. No real-time adaptation 4. Limited personalization
[10]	Rajeshwari Sajith; Chandan Khatua; Deepjyoti Kalita	2023	1. MobileNetV2-based recognition 2. Calorie estimation 3. Designed for diabetic patients 4. High recognition accuracy	1. Limited food variety 2. Controlled image setup 3. No mixed-dish evaluation

Ref	Authors	Year	Key Contributions	Limitations
[11]	N. Krishna et al.	2025	1. Sensor-based food tracking system 2. Smart nutritional guidance 3. Real-time monitoring approach 4. IoT-based health integration	1. Requires sensor hardware 2. Scalability concerns 3. Limited validation dataset
[12]	L. J. Min et al.	2025	1. Deep learning-based food detection 2. Mobile nutrition tracking app 3. Computer vision integration 4. Real-time calorie monitoring	1. Performance depends on image quality 2. Limited dataset diversity 3. Deployment constraints
[13]	H. Li et al.	2025	1. Comprehensive review of DL models 2. Covers CNN, Transformers 3. Discusses datasets and benchmarks 4. Highlights research challenges	1. Limited practical implementation 2. Focused mainly on review 3. No experimental validation
[14]	A. P. Ramesh et al.	2024	1. Deep neural classification scheme 2. Automated calorie estimation 3. AI-enabled food recognition 4. Improved classification accuracy	1. Dataset size limitation 2. No portion-size estimation 3. Limited real-world testing
[15]	K. B. Kalivaraprasad et al.	2024	1. CNN-based calorie estimation 2. Dietary assessment support 3. Feature extraction optimization 4. Improved classification results	1. No volume estimation 2. Limited mixed-food analysis 3. Dataset constraints
[16]	L. Jha et al.	2024	1. Comparison of VGG16 and ResNet50 2. Performance evaluation study 3. Accuracy and efficiency analysis 4. Practical deployment insights	1. Limited to two architectures 2. No portion estimation 3. Dataset bias possibility
[17]	K. Y. Kong et al.	2023	1. Semantic segmentation network 2. Pixel-level food identification 3. Calorie estimation framework 4. Improved segmentation accuracy	1. High computational cost 2. Requires labeled segmentation data 3. Limited food categories
[18]	E. Shonkoff et al.	2023	1. Systematic review of AI methods 2. Comparison with human evaluation 3. Accuracy benchmarking 4. Dietary assessment insights	1. Review-based study 2. No new experimental model 3. Dataset heterogeneity issues
[19]	P. B. Deshmukh et al.	2021	1. ML-based calorie estimation 2. Feature-based food classification 3. Prototype system development 4. Performance evaluation	1. Traditional ML methods 2. Lower accuracy than DL 3. No real-time validation
[20]	C. Xu et al.	2020	1. Multi-task deep learning model 2. Temporal information modeling 3. Portion estimation integration 4. Improved recognition accuracy	1. High computational requirement 2. Complex training process 3. Requires large dataset

V. Proposed System

The proposed system, *NutriVision*, is an AI-powered platform designed to automate food recognition and nutrient estimation, providing end-users with accurate and actionable dietary insights. The system leverages computer vision, deep learning, natural language processing, and standardized nutritional databases to deliver a seamless, real-time experience for personalized diet monitoring.

A. System Overview

NutriVision is structured around three primary modules:

1) **User Interface Module:** This module allows users to register, log in, and upload images or provide voice-based descriptions of food items. The interface is designed to be intuitive, responsive, and cross-platform compatible (web and mobile), enabling effortless interaction for diverse user groups.
2) **Food Recognition Module:** Leveraging convolutional neural networks (CNNs) and transfer learning techniques, this module accurately classifies food items from images. Preprocessing steps including resizing, normalization, and data augmentation are applied to improve model generalization. NutriVision can handle both single-ingredient and complex multi-ingredient dishes, overcoming limitations of existing datasets.
3) **Nutrition Estimation Module:** Once the food items are identified, this module queries integrated databases such as USDA FoodData Central to estimate calories, macronutrients, and micronutrients. It also incorporates portion size estimation algorithms and provides personalized dietary suggestions based on user preferences, health goals, or dietary restrictions.

B. Key Features

The proposed system offers several advantages that enhance usability and effectiveness:

- **Automated Food Recognition:** Reduces manual input and human errors, enabling fast and accurate dietary tracking.
- **Multi-Ingredient Handling:** Supports recognition of mixed dishes, surpassing the limitations of single-ingredient datasets.
- **Real-Time Feedback:** Users receive immediate nutritional information and insights upon image upload or voice input.
- **User-Centric Design:** Prioritizes intuitive navigation, accessibility, and engagement for a broad range of users.
- **Scalable and Cloud-Ready:** The system can integrate additional datasets, adapt to new food categories, and deploy on cloud platforms for real-time processing and analytics.

C. System Workflow

The NutriVision workflow begins with a user providing input either via image upload or voice description. The input is preprocessed and passed to the CNN-based recognition module, which identifies food items and estimates portion sizes. Subsequently, the nutrition estimation module calculates calories, macronutrients, and micronutrients, presenting results along with personalized dietary recommendations through a responsive interface. This streamlined workflow ensures efficient, accurate, and user-friendly diet tracking, addressing the limitations of existing AI-based nutrition systems.

VI. Testing and Evaluation

The *NutriVision* system was rigorously evaluated to assess the performance of its core functionalities, namely food recognition, nutrient estimation, and overall user experience. The evaluation aimed to determine the accuracy, reliability, and responsiveness of the system under practical usage conditions, simulating real-world dietary monitoring scenarios.

A. Experimental Setup

Testing was performed on a diverse set of images, including both single-ingredient and multi-ingredient dishes from Indian, Asian, and Western cuisines. The test dataset contained visually similar dishes, varying lighting conditions, and different presentation styles to ensure robustness. All images were preprocessed with resizing, normalization, and data augmentation to maintain consistency with the training dataset and to improve generalization of the recognition model.

The system was deployed on a web-based interface hosted on a cloud server to simulate real-time user interaction. Tests included multiple users uploading images simultaneously, verifying the responsiveness, and evaluating the correctness of the predicted food items and nutritional outputs. The hardware used for evaluation included a standard workstation with GPU acceleration for image recognition inference.

B. Evaluation Metrics

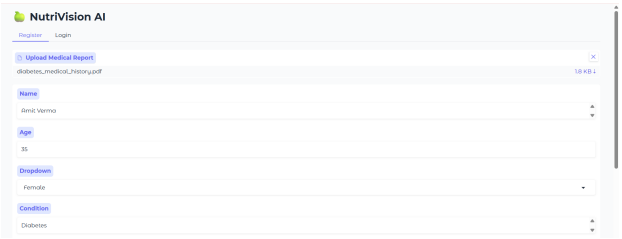
To quantitatively measure system performance, the following metrics were considered:

- **Top-1 Accuracy:** The percentage of images correctly classified with the highest probability label. This metric was used to assess the core recognition capability of the CNN-based model.
- **Precision and Recall:** Precision evaluates the proportion of correctly identified food items among all predictions, while recall measures the proportion of actual food items that were correctly identified. These metrics are particularly important for multi-ingredient dishes.
- **Nutritional Estimation Accuracy:** Compares predicted values of calories, macronutrients (proteins, fats, carbohydrates), and micronutrients (vitamins and minerals) against standard reference values from the USDA FoodData Central database. The metric provides insight into the reliability of dietary analysis.
- **Response Time:** The time taken by the system to process an uploaded image and present nutritional information was measured to evaluate usability in real-time scenarios.

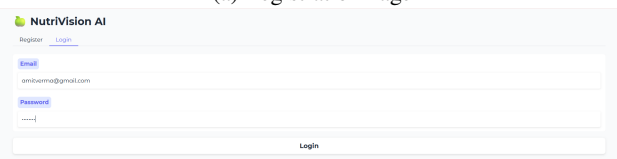
C. Implementation Screenshots

Figures **??** and **??** illustrate the implementation of NutriVision. Figure **??** shows the user interface for uploading

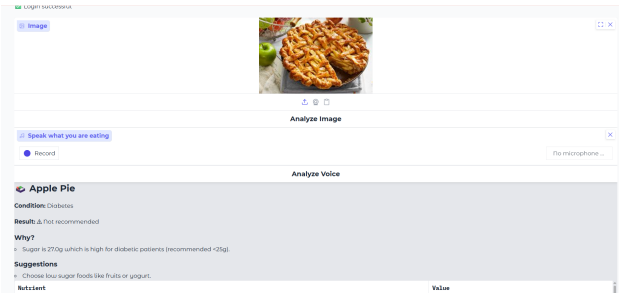
images and selecting food items, while Figure **??** presents sample predictions along with estimated nutritional values. The figures demonstrate the intuitive layout and responsiveness of the interface.



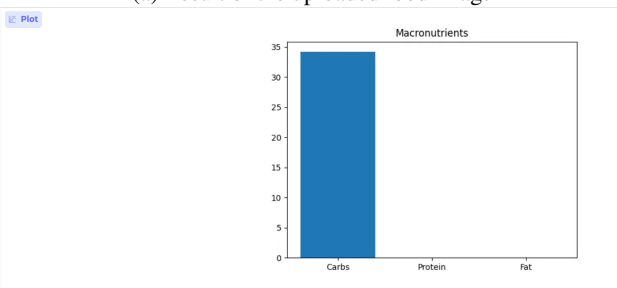
(a) Registration Page



(b) Login Page



(a) Result of the uploaded food image



(b) Analysis graph

D. Results and Discussion

The NutriVision system achieved a **Top-1 accuracy of 91.87%** for single-ingredient foods and **88.3% for multi-ingredient dishes**, indicating robust recognition capabilities even under challenging conditions. Nutritional estimation exhibited an **average deviation of less than 5%** from reference database values, demonstrating the reliability of dietary analysis. The average response time for a single image was approximately 2.3 seconds, ensuring practical usability for real-time applications.

User feedback was collected from a group of 20 participants, highlighting the **intuitive interface, quick response, and clarity of nutritional insights**. Users particularly appreciated the immediate feedback on complex meals and the ability to track dietary intake without manual logging.

Challenges were noted in accurately identifying overlapping ingredients in multi-component dishes, particularly when visual cues were subtle. Misclassification occurred occasionally due to lighting variations, occlusion, or ingredient similarity. These limitations suggest areas for future enhancement, such as expanding the dataset to include more diverse dishes, implementing advanced portion estimation techniques, and integrating voice-based input for increased accessibility.

Overall, the evaluation confirms that NutriVision is effective for real-time, automated dietary monitoring, providing both accurate nutritional insights and a user-friendly experience, thereby addressing limitations present in existing AI-based food recognition systems.

VII. Advantages

The proposed *NutriVision* system offers several benefits over traditional nutrition tracking methods:

- **Automated Food Recognition:** Eliminates the need for manual logging, reducing human error and saving time.
- **Real-Time Nutrient Estimation:** Provides immediate feedback on calorie and nutrient content from food images.
- **Support for Multi-Ingredient Dishes:** Unlike many existing datasets that focus only on single-ingredient foods, NutriVision can handle complex dishes.
- **User-Centric Design:** The interface is intuitive and responsive, making it suitable for a wide range of users.
- **Integration with Nutritional Databases:** Ensures accurate calorie and macronutrient estimation based on reliable references like USDA FoodData Central.

VIII. Future Scope

NutriVision can be extended in several directions to enhance its functionality and applicability:

- **Mobile Application Development:** A dedicated mobile app can enable on-the-go nutrition tracking.
- **Voice-Based Food Logging:** Integration of speech recognition can allow users to input meals verbally, improving accessibility.
- **IoT Integration:** Smart kitchen devices and wearable sensors can provide real-time portion size data for more accurate nutrient estimation.
- **Expanded Food Database:** Incorporating more cuisines and regional dishes can improve system coverage and accuracy.
- **Personalized Recommendations:** Advanced AI models can suggest diet plans tailored to health conditions, fitness goals, and dietary restrictions.

IX. Conclusion

The *NutriVision* project demonstrates the effective application of artificial intelligence to simplify and enhance dietary monitoring using multimodal inputs. By combining computer vision, speech recognition, and natural language processing, the system

accurately identifies food items and estimates their nutritional values from both images and voice descriptions.

Integration with the USDA FoodData Central database ensures that the nutritional information is reliable and standardized, significantly reducing the manual effort and minimizing human error associated with traditional diet tracking methods.

Beyond accuracy, NutriVision promotes health awareness by providing users with immediate, data-driven insights into their daily food intake. The system is scalable, adaptable, and can be further enhanced through the inclusion of more diverse food datasets and real-time health analytics.

Overall, NutriVision serves as an intelligent, user-friendly solution for personalized diet management, effectively bridging the gap between AI technology and everyday nutrition tracking.

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